

Technique 2: Approximation

MP

Teach You Physics Hand In Hand

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1 Solution to Problem in dx-dt Post

- Level of difficulty: **4/10**.
- Classical model: use reduced mass for relative motion.

a) Reduced mass $\mu = \frac{m}{2}$ **2 points**, Coulomb force:

$$\mu \ddot{r} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} \quad \Rightarrow \quad \boxed{\ddot{r} = \frac{q^2}{2\pi\epsilon_0 m} \frac{1}{r^2}} \quad \text{2 points}$$

b) Energy conservation:

$$\frac{m}{4} \dot{r}^2 = \frac{q^2}{4\pi\epsilon_0} \left(\frac{1}{r_0} - \frac{1}{r} \right) \quad \text{8 points}$$

$$t = \sqrt{\frac{\pi\epsilon_0 m}{q^2}} \int_{r_0}^L \frac{dr}{\sqrt{\frac{1}{r_0} - \frac{1}{r}}} \quad \text{7 points}$$

Substitute $r = 1/u$:

$$t = \sqrt{\frac{\pi\epsilon_0 m}{q^2}} \int_{\frac{1}{r_0}}^{\frac{1}{L}} \frac{-du}{u^2 \sqrt{a-u}}$$

Using $\int \frac{du}{u^2 \sqrt{a-u}} = -\frac{\sqrt{a-u}}{au} - \frac{1}{2a^{3/2}} \ln \left| \frac{\sqrt{a} + \sqrt{a-u}}{\sqrt{a} - \sqrt{a-u}} \right| + C$,

$$\Rightarrow t = \sqrt{\frac{\pi\epsilon_0 m}{q^2}} \left[\sqrt{r_0 L (L - r_0)} + \frac{r_0^{3/2}}{2} \ln \left(\frac{1 + \sqrt{1 - r_0/L}}{1 - \sqrt{1 - r_0/L}} \right) \right] \quad \text{1 points}$$

2 Small Number Approximations

- For $a \gg b$, we have $a + b \approx a - b \approx a$.
- For $x \ll 1$, we have $x^n \approx 0$ for $n > 1$.

3 Taylor Series

The Taylor Series gives an approximation of a function $f(x)$ (Never try to remember this thing!):

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Theoretically, for any mathematical expressions in physics problems, we can use Taylor Series to expand and approximate to get a simpler form. However, we only have to use this technique when we face small quantities.

3.1 Small Angle Approximation

Expanding trigonometric functions with small angles with Taylor Series gives:

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \dots \approx \theta$$

$$\cos \theta = 1 - \frac{\theta^2}{2!} + \dots \approx 1$$

$$\tan \theta \approx \sin \theta \approx \theta$$

Since we assume $\theta \approx 0$, any terms involving θ^n where $n > 1$ and be assumed to be 0. We often use this when handling simple harmonic motion.

3.2 Binomial Approximation

By Taylor Series or binomial expansion:

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

For small x , any terms involving x^n where $n > 1$ can be approximated to be 0. Therefore:

$$(1 + x)^n \approx 1 + nx$$

We often use this technique when handling problems involving inverse-square law.

4 Examples

Example 1: Consider a pendulum in a uniform electric field E . A ball with mass m and charge q is attached to a string of length l and the string is suspended from the ceiling. Gravitational acceleration is g . The pendulum starts to swing with small angle. The electric force always oppose the motion of the ball due to manual control. Find the equation of motion of the ball.

Answer 1: Level of difficulty: **2/10**

Assume small angle, so the motion is simply a ball moving rightward or leftward. Vertical force is balanced. Horizontal force:

$$F = ma = T \sin \theta - qE$$

Using small angle approximation with vertical force:

$$T \cos \theta = mg \Rightarrow T(1) = mg \Rightarrow T \approx mg$$

We have $a = \ddot{x} \approx l\ddot{\theta}$, therefore the equation of motion is:

$$\Rightarrow ml\ddot{\theta} = mg\theta - qE$$

This is an ordinary differential equation that we can try to solve.

Example 2: Consider 2 charges of charge $+Q$ placed on x-axis at point $(a, 0)$ and $(-a, 0)$. Another small test charge of charge $+q$ and mass m is displaced from the origin with a small displacement x along the x-axis, where $|x| \ll a$. Find the period of oscillation of the charge q .

Answer 2: Level of difficulty: **3/10**

No vertical force. The horizontal force is given by:

$$F = m\ddot{x} = \frac{qQ}{4\pi\epsilon_0(a+x)^2} - \frac{qQ}{4\pi\epsilon_0(a-x)^2}$$

$$m\ddot{x} = \frac{qQ}{4\pi\epsilon_0 a^2} \left[\left(1 + \frac{x}{a}\right)^{-2} - \left(1 - \frac{x}{a}\right)^{-2} \right]$$

we have $\frac{x}{a} \ll 1$. Using binomial approximation:

$$m\ddot{x} = \frac{qQ}{4\pi\epsilon_0 a^2} \left[\left(1 - \frac{2x}{a}\right) - \left(1 + \frac{2x}{a}\right) \right]$$

$$m\ddot{x} = -\frac{qQ}{\pi\epsilon_0 a^3} x$$

Therefore, period is:

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m\pi\epsilon_0 a^3}{qQ}}$$

5 Problem

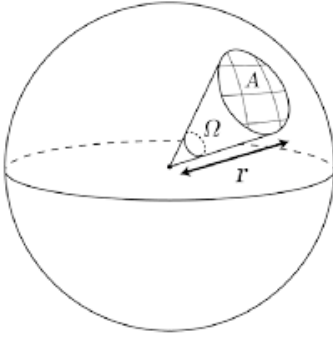
[20 points] In this question, you may use the fact that in spherical coordinate system, the volume of a sphere with radius R is given by:

$$V = \int_0^{2\pi} \int_0^\pi \int_0^R r^2 \sin \theta \, dr \, d\theta \, d\phi$$

Consider 2 planets with mass M located on the x-axis at point $(a, 0)$ and $(-a, 0)$. A small test mass is displaced from the origin along the y-axis with small displacement y and oscillates. Gravitational constant is G .

a)[8 points] Find the period of oscillation.

bi)[2 points] The test mass is a sphere with radius r where $r \ll |y|$. For this sphere, a small angle $d\Omega$ is subtended from a small area dA . By using small angle approximation as a short-cut, prove that the small area dA on the surface of the sphere is $\frac{1}{4}\pi r^2 d\Omega^2$. Also, prove that the circle that encloses the small area dA has a circumference of $\pi r d\Omega$.



bii)[10 points] It is given that for a function $f(\vec{r})$, the Taylor expansion around $\vec{r}_0$ is:

$$f(\vec{r}_0 + \vec{\delta}) \approx f(\vec{r}_0) + \vec{\delta} \cdot \nabla f|_{\vec{r}_0}$$

Assume a tidal force acts on the sphere when the sphere reaches maximum amplitude. Consider an infinitesimal mass element on the surface of the sphere, the tidal force it experiences is:

$$dF = dm \left[\vec{g}(\vec{r}_{\text{center}} + \vec{\delta}) - \vec{g}(\vec{r}_{\text{center}}) \right]$$

Where $\vec{\delta}$ is the displacement vector of the infinitesimal mass from the center. Find the tidal stress (force per unit area) on the sphere.

The solution will be uploaded in the next post and MP love you all :)